

Flying Wing UAV

June 28th, 2026

1 Chuck Glider

I have never designed any sort of wing before, so to avoid getting too ahead of myself and thus never making anything, both Casey and I thought it was best that we make our own little 3-D printed chuck gliders to just test out the process and get used to designing wings.

To start the process of designing the wings, we must look at what we call the design points. These are the points on the wing that determine the fundamental characteristics of the wing behavior. For a chuck gliders whose mission is to stay afloat as long as possible, mimicking one of the core objectives of the surveillance drone, we can find which fundamental characteristic of the wing needs to be optimized first for the mission.

The two starting points at which we start designing our wing are the aspect ratio AR , b^2/S , and the wing load W/S . We start from these points since AR sets our aerodynamic ceiling, the dominant lever on $(L/D)_{max}$ and $C_L^{3/2}/C_D$, also known as how good your glide can be. W/S on the other hand sets where on the curve you will actually fly, fixing your sink and speed rate. One determines the quality of the glide while the other determines the pace of it, everything downstream from here are just realizing the target that you set with these two.

To find which one of these to optimize first depending on our mission we have to do some math.

1.1 AR , Wing Loading, and What Each Mission Optimizes

Setup: Steady Flight We have an unpowered glider descending in a straight line at glide angle γ below horizontal, at airspeed V along the flight path. Three forces: weight W down, lift L perpendicular to the path, drag D along the path opposing the motion. A steady flight implies that the forces stay perpendicular and parallel to the flight path:

$$L = W \cos \gamma \quad D = W \sin \gamma$$

Dividing we get:

$$\tan \gamma = \frac{D}{L}$$

Distance: glide ratio

Glide ratio is horizontal distance per height lost, which is:

$$\frac{1}{\tan \gamma} = \frac{L}{D}$$

Now we substitute in the aerodynamic definitions

$$L = \frac{1}{2}\rho V^2 S C_L \quad D = \frac{1}{2}\rho V^2 S C_D$$

$$\frac{L}{D} = \frac{C_L}{C_D}$$

Notice how the term $\frac{1}{2}\rho V^2 S$ cancel. Resulting in the glide ratio depending only on the **drag polar**, no ρ , no W , no S .

Distance covered from height h is just:

$$h \cdot \left(\frac{L}{D}\right)_{max}$$

The glide ratio, which is dimensionless ratio that is described as horizontal distance per unit height, get multiplied by height to then get the net horizontal distance traveled.

Now how does the $\left(\frac{L}{D}\right)_{max}$ relate to either of our fundamental controlled variables, AR and W/S . Well we already established above that the W and S components cancel so that leaves us with only AR. Recall the equation for C_D :

$$C_D = C_{D_0} + \frac{C_L^2}{\pi e AR}$$

Parasite drag being C_{D_0} , which is a constant set by the wetted area and surface quality (surface friction) plus induced drag which carries our AR term that we are interested in. To simplify further calculations we set $k = \frac{1}{\pi e AR}$ to get $C_D = C_{D_0} + kC_L^2$.

Now to maximize $\frac{L}{D} = \frac{C_L}{C_{D_0} + kC_L^2}$ we set the derivative to 0:

$$\frac{d}{dC_L} \left(\frac{C_L}{C_{D_0} + kC_L^2} \right) = 0 \rightarrow C_{D_0} - kC_L^2 = 0$$

The physical interpretation of this is that the best glide happens when the parasitic drag and induced drag equal each other. The shared point of the two halves of the polar balance.

Solving gives us $C_L = \sqrt{\frac{C_{D_0}}{k}}$, subbing back in the total drag is $C_D = 2C_{D_0}$ so:

$$\left(\frac{L}{D}\right)_{max} = \frac{1}{2\sqrt{kC_{D_0}}} = \frac{1}{2}\sqrt{\frac{\pi e AR}{C_{D_0}}}$$

We can see that our $(\frac{L}{D})_{max}$ ceiling is determined by $\sqrt{AR}$, note that AR has diminishing returns quadrupling AR only doubles $\frac{L}{D}$.

This at first glance might make you think that optimizing AR simply requires cranking AR as high as possible until the structure can't manage it but you must notice that e and C_{D_0} (for our low Re regime) also get effected by AR.

Oswald's efficiency factor, e , falls as AR rises which eats into our $\sqrt{AR}$.

On top of that a higher AR at a fixed area gives us a shorter chord $\rightarrow$ lower chord Reynolds number $\rightarrow$ thicker separation bubbles $\rightarrow$ higher C_{D_0} which sits underneath in our $(\frac{L}{D})_{max}$.

So as our AR rises we see three effects compound: our ceiling only climbs due to our $\sqrt{AR}$ which gives diminishing returns, the e factor falls which eats into our $\sqrt{AR}$, all while the floor C_{D_0} rises through the shrinking chord. The gains slow just as the penalty accelerates.

Our optimal AR is set by the Reynolds floor not the structure, we will go more into this when we cover Reynolds number for our glider.

Endurance: height over sink rate

Our time afloat descending from a give height h is: $t = \frac{h}{V_s}$, so maximizing our endurance means minimizing sink rate V_s . We define our sink rate is the vertical component of the flight path speed:

$$V_s = V \sin \gamma \approx V \cdot \frac{C_D}{C_L}$$

Breaking this simplification down we have two separate moves. The first one sets $V_s = V \sin \gamma$, remember that $V \sin \gamma$ is the vertical component and $V \cos \gamma$ is the horizontal component. We are trying to figure out how to stay afloat the longest, ie how much height is being lost, straight down. If $\gamma = 0$ the glider wouldn't sink at all and if $\gamma = 90$ the entire velocity would be going straight down. Now our second move introduces aerodynamics where we set $\sin \gamma \approx \frac{C_D}{C_L}$. From the setup we have $\tan \gamma = \frac{C_D}{C_L}$ and for small angles $\tan \gamma \approx \sin \gamma$. We assume that the angle of descent is small as this should be the case on any decent glider.

Now after our initial setup for endurance we bring in speed from $L = W \cos \gamma \approx W$ (we use the same small angle approximation $\cos \gamma \approx 1$). This allows us to set $L \approx W$.

$$L = \frac{1}{2}\rho V^2 SC_L = W$$

$$V^2 = \frac{W}{\frac{1}{2}\rho S C_L} = \frac{2}{\rho C_L} \cdot \frac{W}{S}$$

$$V = \sqrt{\frac{2(W/S)}{\rho C_L}}$$

Now substitute back into V_s

$$V_s = \sqrt{\frac{2(W/S)}{\rho}} \cdot \frac{C_D}{C_L^{3/2}}$$

In distance we saw W , S , and ρ cancel out, in endurance this is not the case. But we also still have a polar left which we want to maximize just like we did for distance, we want $\frac{C_L^{3/2}}{C_D}$ to be as large as possible.

So we maximize $\frac{C_L^{3/2}}{C_{D_0} + kC_L^2}$:

$$\frac{3}{2}C_{D_0} - \frac{1}{2}kC_L^2 = 0 \rightarrow kC_L^2 = 3C_{D_0}$$

Recall that the best glide is when parasite drag was equal to induced drag:

$$kC_{L_{glide}}^2 = C_{D_0}$$

Here the minimum sink is where induced drag is 3 times parasite:

$$kC_{L_{sink}}^2 = 3C_{D_0}$$

Which gives

$$C_{L_{sink}} = \sqrt{3}C_{L_{glide}}$$

Switching into terms of speed V we recall that $V = \sqrt{\frac{2(W/S)}{\rho C_L}}$ to get:

$$\frac{V_{sink}}{V_{glide}} = (\sqrt{3})^{-1/2} = (3)^{-1/4} \approx 0.76$$

This directly tells us that to get the best loiter we need to slow our glider down by 24 percent from ideal glide speed if we want to maximize our endurance.

Now if we sub C_L back we get $C_D = 4C_{D_0}$ and:

$$\left(\frac{C_L^{3/2}}{C_D}\right)_{max} \propto \frac{AR^{3/4}}{C_{D_0}^{1/4}}$$

If we compare this back to our distance:

$$\left(\frac{L}{D}\right)_{max} \propto \frac{AR^{1/2}}{C_{D_0}^{1/2}}$$

We see that endurance leans harder on AR ($AR^{3/4}$ vs $AR^{1/2}$) than distance but softer on the parasite floor ($C_{D_0}^{1/4}$ vs $C_{D_0}^{1/2}$). Which as stated in distance leads us even harder into the low-Re trap which a high AR creates at low Re.

We found that endurance relies more heavily on AR than distance does, but it also relies on W/S which cancels out in distance. **W/S does nothing for distance and effects everything for endurance**, which draws us to the conclusion that tuning W/S first is the correct decision for building an aircraft whose mission is to loiter as long as possible. You want a high AR for both missions, W/S distinguishes between the two.

An analogy that I feel sums it up well:

Both endurance and distance missions spend in the same currency, the starting height h . The differ in where they convert that currency: distance converts h into horizontal reach through $(\frac{L}{D})_{max}$, endurance converts h into time through $\frac{1}{V_{s,min}}$. AR improves the exchange rate for both; W/S is the tie breaker that says which conversion you are optimizing

1.2 Picking our fundamental design points

As determined in 1.1 we need determined that selecting wing loading first for our given mission, having the chuck glider loiter as long as possible, as the design point to pick.

This is pretty intuitive (just make the aircraft as light as possible) but the derivations helped me get my head sort of back in the game and formalized everything moving forward giving me a good starting off point to elaborate from.

I need to take a step back, I cannot directly pick wing loading. I need to look at what I am being limited by: my printer size.

The bed of my printer is 220mm which doesn't limit necessarily the direct size of my wing (I can join pieces together) but limits the chord which I need to be able to print in one piece. I can join pieces along the span but not along the chord. This is my true starting off point.

The derivations above rather than serving as a jumping off point establishes W/S as the point in which the endurance objective checks achieves. My printer geometry couples AR and W/S, so we will reach the endurance objective indirectly.

Instead of picking a target endurance, like we would with the main drone, which could then lead to an optimal AR and W/S to achieve the target time in the air.

We start by setting what we are limited by:

- Root chord (c_{root}): 22cm

1.3 The printer picks our wing loading

After doing some more digging I realize that the printer does only just limit the chord but also limits our wing loading as well. We are printing in ABS since I would like to experiment using acetone to both smooth the glider but also join the pieces together.

A printed wing is essentially two skins, a top and a bottom surface, wrapped around some internal structure. The weight of those skins per unit of plan area is:

$$w_{skin} \approx 2 \cdot \rho_{ABS} \cdot t_{wall} \cdot 1.05$$

The factor of 2 counts both surfaces, t_{wall} is the wall thickness (perimeters $\times$ line width, so one 0.45mm perimeter gives $t_{wall} = 0.45\text{mm}$). I arrived on 0.45mm as this is the thinnest my printer can reasonably print, if I need to backtrack if I realize in the CAD that it is too thin then that is just what I will have to do.

And the 1.05 accounts for the fact that the curved surface of the airfoil is slightly longer than the chord it wraps. With $\rho_{ABS} \approx 1040 \text{ kg/m}^3$ and a single wall:

$$w_{skin} = 2(1040)(0.00045)(1.05) \approx 0.98 \text{ kg/m}^2$$

The skins are not the whole story. Adding roughly 25 percent for the spar, the joint faces, and the solid leading and trailing edge sections gives a structural areal weight of:

$$w_{areal} \approx 1.2 \text{ kg/m}^2$$

On top of the structure, a flying wing needs nose ballast to place the CG, which I will write as a fraction β of the structure weight. Until I do the actual CG budget I will assume $\beta \approx 0.3\text{--}0.5$.

Now the total weight of the glider is just this areal weight times the wing area:

$$W = w_{areal}(1 + \beta) \cdot S \cdot g$$

Dividing both sides by S:

$$\frac{W}{S} = w_{areal}(1 + \beta) \cdot g$$

The area cancels. Making the glider bigger or smaller does not change the wing loading at all. W/S is fixed entirely by the material density, the number of perimeters, and the ballast fraction. In other words, **my printer settings are my wing loading selection**. In 1.1 we concluded that W/S is the design point to tune first for an endurance mission; it turns out that for a printed aircraft that tuning happens in the slicer, not in the geometry. Plugging in the numbers:

$$\frac{W}{S} = 1.2 \cdot 1.4 \cdot 9.81 \approx 16.5 \text{ N/m}^2 (\approx 1.68 \text{ kg/m}^2)$$

1.3.1 The cost of ABS

This is where the material choice bites. Recall from the endurance derivation that sink rate scales with the square root of wing loading:

$$V_s \propto \sqrt{W/S}$$

LW-PLA foams to roughly half the effective density of ABS, so an ABS wing carries about twice the areal weight, which costs roughly 40 percent more sink rate ($\sqrt{2} \approx 1.4$). I am accepting that penalty deliberately: ABS is far cheaper, and since my printer only lets me join pieces along the span, every spanwise joint is structural. Acetone welding ABS produces an essentially 'factory sealed' joint, stronger than any glue joint I could make in PLA. For a test article whose mission is teaching me the process (and that will be thrown at the ground repeatedly), cheap and tough beats optimal.

1.3.2 Updated constraints

The printer and material therefore hand me two of my fundamental quantities directly:

- Chord (c): $\leq 22\text{cm}$, set by the print bed
- Wing loading (W/S): $\approx 1.68 \text{ kg/m}^2 \approx 16.5 \text{ N/m}^2$, set by ABS density, wall count, and ballast fraction of $\beta = 0.4$

With W/S locked, the geometry no longer controls the wing loading at all. What it does control is the chord Reynolds number and the aspect ratio, which is exactly where the low-Re trap from 1.1 lives. That sizing loop is the next section.

1.3.3 Realizing the complexity in 3D printing for our specific mission

Before I moved onto the next section I realized that half a millimeter is unreasonably thin and the glider hitting the ground nose first would not end well with such thin walls. I ran the calculations again if the walls were thickened to 0.9mm (I also look at if certain parts of the aircraft would be reinforced, not only the entire thing) and the W/S quickly rose. The mission for the flying wing glider is for it to stay afloat as long as possible and this wrinkle if you will makes said mission give poor results.

On top of this the 3D printer is not mine and printing in ABS requires you to cover your bed to keep a constant temperature assuring that the filament will not crack. The solution I would be able to give to this problem without having to add any permanent modifications would just be placing a cardboard box over the printer; this is quite sketchy and I'm not even sure it would produce the best results. The reason I selected ABS is its ability to be smoothed and glued with acetone and it being lighter PLA.

1.4 Alternate fabrication process: using a hot wire

My solution to this after some googling it to make a hot wire setup to cut the wing out of foam. The difficulty in this task of course is making sure the cuts are near pinpoint accurate. I found a guide online specifically showing how you could create a gizmo that pulls your hot wire consistently along the foam allowing me to create a tapered wing.

1.4.1 The benefits and downsides of using hot wire and foam

There are three large benefits of using a hot wire to make a wing:

-1 The density of XPS foam (specifically Owens Corning) is significantly lighter than ABS, coming in at 25 kg/m^3 vs $\approx 1040 \text{ kg/m}^3$. Of course the foam wing would be filled and the ABS would have a thin shell of about 1mm but the difference is still huge we can run an approximation of the difference assuming a 22cm chord and a 10 percent thick flying section:

$$A \approx 0.685 \cdot (t/c) \cdot c^2 = 0.685 \times 10 \times 0.22^2 \approx 33\text{cm}^2 (0.0033\text{m}^2)$$

The 0.685 is the airfoil 'fullness' factor (a rectangle would be 1), the number comes from NACA 4 digit symmetric section.

$$P \approx 2.05 \cdot c \approx 0.45\text{m}$$

The 2.05 being the perimeter factor which I got from the NACA 4 digit airfoil data base.

So comparing ABS vs foam:

- Solid ABS: $0.0033 \times 1040 \approx 3.45 \text{ kg/m}$
- Hollow ABS (1mm wall): $0.45 \times 0.001 = 0.00045 \text{ m}^2$, $\times 1040 \approx 0.47 \text{ kg/m}$ (470 g/m)
- Solid foam core: $0.0033 \times 25 \approx 0.083 \text{ kg/m}$ (83 g/m)

For the foam we might need some rods going through it for support we will see through experimentation and definitely will need tape as a film over the foam but at most that will add 30-40 g/m (since the large majority of the wing will just be foam). This still makes foam 4x lighter than the ABS that is just a thing wall with no joints, ribs, etc.

Remember that the main mission of the glider is to stay afloat for as long as possible, W/S being the main knob that controls this, so the foam is much better suited for this job.

-2 A chuck glider will inevitably have is fair share of rough landings and crashes. The foam is able to be repaired significantly more easily with glue, tape, and leftover foam. ABS if it cracks will be significantly harder to repair if not impossible and will require a new print of the section. For an aircraft that I predict will crash often this matters enough to give a serious edge to foam.

-3 Perhaps the greatest advantage is the speed I will be able to reiterate designs with a fully assembled hot wire setup.

Designing and printing a new wing will at the quickest require an overnight print (even more if it needs to be printed in many sections that need to be taken in and out).

Using a hot wire setup once proficient will take a an hour at most to cut out the wings with significantly less cost in foam compared to filament.

This is the difference between being able to prototype 5 wing variations in a month vs 1-2.

On top of this we could potentially use the foam as a mold for the wings for when we want to 'cast' the final drone out of carbon fiber. I don't think we would be able to use foam for all parts of the drone like the fuselage where CADing the parts where our payloads will go will be much more convenient and more precise then hand carving foam, but the setup could still be used.

The two main disadvantages of using foam:

-1 The largest disadvantage of using a hot wire setup that is not on four axes (these cost a couple thousand dollars) is the inability to create nonlinear twist. I think an entire subsection could honestly be dedicated to why twist matters for creating flying wings specifically but from my limited googling it is not the only way you can make flying wings, and honestly the more well established method of making flying wings is using re-flexed airfoils which a hot wire setup is capable of making.

-2 The geometric fidelity is significantly more shaky, meaning the design that you have in mind of creating is not necessarily going to be 1 to 1 like 3D printing would achieve. With enough practice and with the correct setup I am pretty confident that I could manufacture wings with accuracy but this would take time and money (foam). The question is whether the time it would take to learn how to use the hot wire setup would payoff.

The verdict (well not really):

The main point pulling me in the direction of using a hot wire setup is the speed at which would I would be able to reiterate designs in the future. This test chuck glider is only a small part of this project so investing into a hot wire setup only makes sense if I could keep using the setup. From the research I have done it seems that this is the case and has major benefits once you have gotten enough practice. I think for making a project of this scale at least a couple iterations of prototypes would be necessary and if manufacturing these iterations can be made streamlined investing some money would be acceptable. I also think it would be fun to make my own wings 'by hand' and making the hot wire setup in the first place.

The main limiter it seems that the setup impedes is only being able to make a reflexed flying wing cutting out the possibility of making one with nonlinear

twist. Before I can make a proper verdict on whether investing into a hot wire setup is worth it I need to do research on this as well as figure out how much the setup would cost in the first place.

As I mentioned before the printer is not mine which is significant if I need to make many prints not to mention that printing in ABS I would need to make some sketchy temporary changes to the printer.

Right now I am leaning towards doing the hot wire but I still need to do some more research before making a final decision.

<https://www.rcgroups.com/forums/showatt.php?attachmentid=17685907>

1.4.2 The process in creating the hot wire assembly

Materials needed:

The cutting bow

- Nichrome resistance wire - Heated cutting element - $\approx$ 8 dollars
- Hardwood handle or pvc pipe - handle - free
- Spring-steel wire - Holds the nichrome wire taught - $\approx$ 6 dollars
- Alligator clamps - connects power to each end of the nichrome wire - free
- Heat shrink tubing - insulates the areas where I solder - free

Electrical

- Switch - turn on and off power - $\approx$ 5 dollars
- Silicone hookup wire, 16–18 AWG - carries current from supply to bow - $\approx$ 6 dollars
- Adjustable DC bench supply - adjustable power to the wire this is important when some parts need to be cut with more delicacy - free (I can borrow one for the summer)

Templates

- Plywood - the material that I will build the root and tip templates out of - free
- Spray adhesive - to put paper template on plywood to get accurate cut - free
- Sandpaper - smooth out the edges for an accurate template - free
- Small nails - label my templates - free

Foam

- XPS insulation foam (1.5in if you can find it) - the wing core - 25 dollars a sheet (one sheet can make many wings)

Gizmo that lets me cut accurately

- Two 52 in pine stringers (fixed + swing arm), four $1\frac{1}{4}$ in pulleys, a bushing + bolt pivot, braided line, three spring clamps, a 1 lb weight — automates a synchronized tapered cut — 30 total

Total:

80 dollars with enough foam to make ≈ 8 wings

1.4.3 Reflexed airfoil vs nonlinear twist

I will start with the core definitions needed before going forward:

Twist:

Twist is any spanwise variation in the angle at which the wing sections meet the flow. What actually matters aerodynamically is the local effective angle of attack, measured from the section's zero-lift line:

$$\alpha_{eff}(y) = \alpha_{geo}(y) - \alpha_{L0}(y)$$

Twist is how you make α_{eff} vary along the span, and it breaks down into two parts:

Geometric twist physically rotates the chord line from station to station. If the tip sits at a lower incidence than the root, that is **washout**; the opposite is **washin**. This is the most common definition of 'twist.'

Aerodynamic twist changes α_{L0} instead of the chord angle, by changing the airfoil selection spanwise (typically camber). A more cambered section has a more-negative α_{L0} , so even with no geometric twist the effective angle differs. Reflex is how we manipulate this.

The power manipulating twist and reflex can be seen when we do some math:

Assumptions:

Standard lifting line theory (TAT in 3D essentially): high aspect ratio so each section behaves locally in 2D, small angles, linear section lift curve, and a single bound vortex line and a quarter chord flat trailing wake. Semi-span s , coordinate $y \in [-s, s]$.

Two expression for local lift: -1 Kutta-Joukowski. The bound circulation $\Gamma(y)$ gives lift per unit span $L'(y) = \rho V_\infty \Gamma(y)$. Same thing but with section lift coefficient $L'(y) = \frac{1}{2} \rho V_\infty^2 c(y) c_l(y)$ remember:

$$c_l(y) = \frac{2\Gamma(y)}{V_\infty c(y)}$$

$c(y)$, this is chord, is the conversion between circulation and local c_l

-2 Section theory. Locally lift follows its 2D lift curve, but the section doesn't see V_∞ ; the trailing wake induces a downwash that tilts the local flow down by the induced angle $\alpha_i(y)$. So the angle the section actually flies gets reduced:

$$c_l(y) = \alpha_0(y)[\alpha_{geo}(y) - \alpha_{L0}(y) - \alpha_i(y)]$$

Remember from twist:

$$\alpha_{eff}(y) = \alpha_{geo}(y) - \alpha_{L0}(y)$$

Here α_0 is the section lift-curve slope (cnst). And α_i is our only unknown.

Downwash

This is tickling a part of my brain I did this in MAE 136!

Our α_i is the term that connects every station to every other one. Since Γ varies along the span, Helmholtz says vorticity of strength $d\Gamma/dy$ is shed into the wake. And by Biot-Savart it induces a station y at downwash

$$w(y) = \frac{1}{4\pi} \int_{-s}^s \frac{d\Gamma/dy_0}{y - y_0} dy_0, \quad \alpha_i(y) = \frac{w(y)}{V_\infty}$$

$w(y)$ is downwash. This is why span load is a global problem, when you set Γ at one station it changes the α_i everywhere else due to the wake that Γ sheds.

So now we can define α_i

Combine:

Setting -1 equal to -2 and subbing in α_i

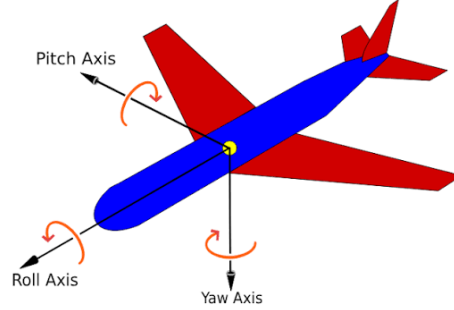
$$\frac{2\Gamma(y)}{V_\infty c(y)} = \alpha_0[\alpha_{geo} - \alpha_{L0}(y) - \frac{1}{4\pi V_\infty} \int_{-s}^s \frac{d\Gamma/dy_0}{y - y_0} dy_0]$$

The only unknown is circulation everything else is geometry. Rearranging where we put the unknowns to the left:

$$\frac{2\Gamma(y)}{V_\infty c(y)\alpha_0(y)} + \frac{1}{4\pi V_\infty} \int_{-s}^s \frac{d\Gamma/dy_0}{y - y_0} dy_0 = \alpha_{geo}(y) - \alpha_{L0}(y)$$

Look at the right hand side, it is solely a function of twist α_{geo} and section α_{L0} , and the linear operator on the left carries the chord $c(y)$. With these three inputs we can control our Γ . This lets us control the lift and specific stations along the wing which allows us to manipulate how the wing stabilizes itself.

Effectively the power of being able to manipulate both the twist and reflex of an airfoil lies in being able to control the local circulation $\Gamma(y) \rightarrow$ lift $L'(y)$. Lift acts at a distance from the CG (center of gravity) of the aircraft which exerts a **rolling moment** ($L'(y) \cdot y$) and if the station sits fore or aft of the CG it also exerts a **pitching moment** ($L'(y) \cdot x$).



What it does *not* do directly is yaw the aircraft: lift is nearly entirely vertical at small angles and therefore as no lever about the vertical axis. Yaw instead comes from the horizontal forces, the spanwise **induced drag** distribution $(D'_i(y) \cdot y)$, which is downstream to the lift distribution not the force itself.

So what does manipulating the twist and reflex buy us?

We get to control the local lift at each station which then allows us to control the roll and pitch of the aircraft.

Yaw is a second-order payoff of the same knob.

Controlling $\Gamma(y)$ does not just set lift, it sets the induced drag, and the induced drag is what actually yaws the aircraft. The local induced drag per span is the lift vector tilted back by the induced angle:

$$D'_i(y) = L'(y) \sin \alpha_i(y) \approx \rho V_\infty \Gamma(y) \cdot \frac{w(y)}{V_\infty} = \rho \Gamma(y) w(y)$$

So $D'_i(y)$ is the *product* of the local circulation and the local downwash. I cannot set it locally: $w(y)$ is the global functional from before, so the *shape* of Γ decides where the drag lives.

This is where elliptic and bell split. Both loadings die to zero at the tips ($\Gamma(\pm s) = 0$ is forced on any finite wing, a free tip cannot hold a pressure jump), so the distinction is *not* the tip lift. It is the sign of the downwash near the tip:

- **Elliptic:** downwash is constant across the span, so $D'_i = \rho w \Gamma$ is positive everywhere. Drag lives across the whole wing.
- **Bell:** the inboard-shifted load gives downwash inboard but **upwash outboard**. Where $w < 0$, $D'_i < 0$: the tips make induced **thrust**, not drag.

The bell is not free. Relocating load inboard costs total induced drag (elliptic is the true minimum); I accept that because the bell is the minimum-drag solution for a given *root bending moment* (lighter structure) and because the thrust at the tips is exactly the lever that fixes yaw.

Proverse yaw. To roll, the elevons out at the tips perturb Γ antisymmetrically. On the bell the tips sit in the induced-thrust region, so adding load at a tip adds *thrust* there, pulling that wing forward and swinging the nose *into* the roll: proverse yaw, coordinated for free. On an elliptic wing the tips are in drag, so the same input drags the wing back and yaws *against* the roll, the classic adverse yaw I would otherwise trim out with a rudder.

”Stability” is two different jobs. On a tailless wing I have to keep these separate:

- **Directional (weathercock) stability** ($C_{n_\beta} > 0$): the nose swinging back after a sideslip. Needs side-force area *aft of the CG*, a fin, winglets, or swept tips.
- **Turn coordination:** the yaw produced *during* a roll (adverse vs proverse). This is the induced-drag effect above.

The bell spanload fixes coordination. By itself it does *not* give weathercock stability: a straight bell wing still wants a fin.

Sweep is what ties them together. With sweep the tips sit *aft* of the CG. Under sideslip the windward tip’s induced drag changes, and because that drag now acts on a lever behind the CG it produces a restoring yaw moment. So **bell + sweep** buys the tailless, finless, directionally stable result (the Prandtl-D). The catch is that sweep makes the outboard washout do **triple duty**: it shapes the bell for yaw, it sets longitudinal trim (swept tips lift behind the CG, i.e. nose-down, so washout is what unloads them), and it governs stall (underwashed swept tips stall first and pitch the nose up). Three jobs riding on the one washout distribution.

Back to the title: reflex vs nonlinear twist. These are my two levers for shaping the load and trimming the wing without a tail:

- **Reflex (aerodynamic twist):** enters the lifting line through $\alpha_{L0}(y)$, and, crucially, *outside* the $\Gamma(y)$ analysis, supplies the nose up C_{m0} that lets a tailless wing trim. On a hot wire it is essentially **free**: it is baked into the template airfoil, so every ruled station inherits it.
- **Nonlinear geometric twist (the bell washout):** shapes $\alpha_{geo}(y)$ directly and is what the true bell requires. On a hot wire this is the **expensive** lever, as a wire is straight thus can only cut a straight line

The hot-wire verdict. A two-template hot wire cuts a *ruled surface*: the wire is a straight line swept between the root and tip templates, so the geometry linearly interpolates between them. A single cut can therefore only give me **linear** twist, not the nonlinear washout the bell wants. This does *not* rule out the bell. Since I can only join along the span anyway, I can cut the semi-span as several panels, each its own linear ruled cut with its own twist, and glue them into a **piecewise-linear approximation** of the bell washout. The cost is that every joint adds hand-cutting error, and it accumulates worst at the tips, which

is exactly where the bell's payoff (the thrust region, and on a swept wing the directional stability itself) is most sensitive.

So the hot wire survives the analysis. It does not force me off the bell; it turns "smooth nonlinear washout in one cut" into "piecewise-linear washout across a few glued panels," which degrades precisely where it matters. For a first article whose real mission is learning the process and surviving crashes, the bell is a **stretch goal, not a day-one cut**. The day-one wing is the hot-wire-native one:

- a **reflexed section** so the wing trims tailless without depending on precise washout, since reflex is free on the template;
- **swept**, so the eventual finless path stays open
- **conservative (generous) linear washout** to keep the tips unloaded (docile stall, reduced adverse yaw), well inside what one clean cut per panel can hit;
- **small winglets** as cheap directional-stability insurance rather than betting it all on a first hand-cut spanload.

Wing two, once my panels come off the wire repeatably, is the paneled piecewise-linear bell where I try to drop the winglets. That is where the elegant Prandtl result shows up, earned rather than attempted cold.

1.5 The new design points to start from

I am no longer limited by my printer size so my chord is not necessarily the design point that I must pick. On top of that my W/S binding from ABS no longer follows since the print relied on the total skin of the airfoil (S) and depth x, we no longer have an S to cancel since our foam fills the whole wing not just the skin.

What am I limited by now:

My wire length binds my span for each given panel I cut. For this first piece however I want to cut my wing in one piece for simplicity.

My wire length is limited by how much length I can put on before it starts to sag. After some googling the RC plane community consensus is from 60-80cm; I will pick 60cm since I want as accurate cuts as possible.

What also binds me is the sheet thickness which translates to the airfoils absolute thickness. The cheapest foam sheets are 1in or 25.4mm, this gives about 22mm of usable thickness.

My bound numbers:

- Span: 60cm, minus a little margin give $\approx 55\text{cm}$ of usable space per panel which gives a 1.1m span $\rightarrow b = 1.1\text{m}$

- Chord: airfoils are not defined by there thickness but by there t/c ratio. After looking at the NACA airfoil database reflexed airfoils for flying wings are in between 8-10 percent, we take the lowest chord possible (10 percent) to get the highest AR possible. $c_{max} = \frac{t_{usable}}{t/c}$. This gives us ironically the same max chord of 22cm. There is a wrinkle I did not consider right away, at 22cm the Re would get dangerously low at the tips, a clause that I had mentioned in 1.1. To get in a safe Re a t/c of 8 percent will be chose instead making the AR slightly lower but putting us in a safe Re tip range. This gives us: $c_{max} = \frac{22mm}{0.08} = 275mm$ (0.275m)

1.5.1 Determining taper

At this point the tooling and material have fixed everything except for one geometric variable. The wire length sets the span ($b = 1.1m$), the sheet thickness with the section thickness ratio sets the root chord ($c_{root} = c_{max} = 0.275m$), and the materials (foam density, covering, and ballast function) sets the areal weight law. The last free ratio is the taper ratio:

$$\lambda = \frac{c_{tip}}{c_{root}}$$

Everything downstream is a function of λ , so choosing it closes the geometry completely.

Geometry as a function of λ . For a linear taper the local chord is

$$c(y) = c_{root} \left[1 - (1 - \lambda) \frac{|y|}{s} \right]$$

which gives the mean chord, area, aspect ratio, and tip chord:

$$\bar{c} = c_{root} \frac{1 + \lambda}{2} \quad S = b \bar{c} \quad AR = \frac{b^2}{S} = \frac{b}{\bar{c}} = \frac{2b}{c_{root}(1 + \lambda)} \quad c_{tip} = \lambda c_{root}$$

Note that AR is decreasing in λ and c_{tip} is increasing in λ . Taper ratio is therefore a trade between aspect ratio and tip chord. Which is the same AR vs Re as the chord selection, now specifically at the tip.

The wing load as a function of λ . The foam core is a volume, so the areal weight varies along the span with local chord:

$$W(y) = \rho_f K (t/c) c(y) + W_{cover}$$

where $K \approx 0.685$ is the section fullness factor (placeholder from the NACA 4-digit family (like in 1.4.1), to be replaced by the chosen section's true value). Integrating over the span and dividing by S replaces $c(y)$ with $\bar{c}$:

$$\frac{W}{S} = g (1 + \beta) [\rho_f K (t/c) \bar{c} + W_{cover}]$$

This is why W/S depends on λ , the mean chord sits inside it. The dependence is weak because the covering term is constant and the core term is linear in $\bar{c}$; across $\lambda \in [0.4, 0.7]$ the wing loading only spans $\approx 4.7\text{--}5.5 \text{ N/m}^2$.

Trim speed and tip Reynolds number. From the endurance derivation, the glider loiters at

$$V(\lambda) = \sqrt{\frac{2(W/S)(\lambda)}{\rho C_L}}$$

evaluated at the minimum-sink C_L . The Reynolds number at the tip, the most starved station, is then

$$Re_{tip}(\lambda) = \frac{V(\lambda) \cdot \lambda c_{root}}{\nu}$$

Re_{tip} rises with λ both directly through c_{tip} and weakly through V . There is no circularity in this chain: for any candidate λ every quantity is fully determined, so the chain $\lambda \rightarrow \bar{c} \rightarrow W/S \rightarrow V \rightarrow Re_{tip}$ is a function evaluation, not a loop.

The selection criterion. Low λ buys aspect ratio and span efficiency but starves the tip of Reynolds number. This matters disproportionately on a swept flying wing: a tip flying below $Re \approx 40\text{k}$ sits in the erratic bubble regime, and a swept wing that stalls at the tip pitches *up*. The washout is also asked to do three jobs at the tip (bell shaping, trim, stall order), all of which presume the section actually delivers its polar there. High λ feeds the tip but spends aspect ratio the endurance mission wants ($AR^{3/4}$). The rule is therefore:

choose the smallest λ whose tip clears the Reynolds floor $Re_{tip} \gtrsim 40\text{k}$

Evaluating the chain at $c_{root} = 27.5 \text{ cm}$, $t/c = 0.08$, $C_L = 0.6$:

Evaluating the chain gets us:

λ	$c_{tip} \text{ (cm)}$	$\bar{c} \text{ (cm)}$	AR	$V \text{ (m/s)}$	Re_{tip}
0.4	11.0	19.3	5.7	3.6	$\approx 28\text{k}$
0.5	13.8	20.6	5.3	3.7	$\approx 35\text{k}$
0.6	16.5	22.0	5.0	3.8	$\approx 42\text{k}$
0.7	19.3	23.4	4.7	3.9	$\approx 49\text{k}$

The floor bites at $\boxed{\lambda \approx 0.6}$: tip $Re \approx 42\text{k}$, $AR = 5.0$, $W/S \approx 5.2 \text{ N/m}^2$. Dropping to $\lambda = 0.5$ buys 0.3 of aspect ratio and puts the tips back in bubble country; rising to 0.7 buys tip margin that is not needed at the price of aspect ratio that is.

Note the table below is exact conditional on three placeholders - β , K , and trim C_L - each of which enters V only under a square root, and each of which gets replaced by a measured/computed value at the airfoil-selection and CG-budget stages without reopening the λ choice.

What this closes. With λ set the planform is fully determined: $S \approx 0.242 \text{ m}^2$, $AR \approx 5.0$, all-up weight $\approx 130 \text{ g}$ (of which $\approx 35\text{--}40 \text{ g}$ is nose ballast at $\beta = 0.4$),

trim speed ≈ 3.8 m/s, root $Re \approx 70k$, tip $Re \approx 42k$. The remaining decisions are sweep angle (directional stability), washout amount (one linear cut), and the airfoil pair, which can now be selected against known station Reynolds numbers rather than guesses.

Don't use the wrong number! The $\bar{c}$ above is the arithmetic mean chord, correct for S , AR , and W/S . The mean *aerodynamic* chord used for CG and neutral-point placement is a different average,

$$MAC = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

(≈ 22.4 cm at $\lambda = 0.6$, located outboard of the centerline). Use the arithmetic mean for anything area- or weight-related and the MAC for anything moment-related.

1.5.2 This irks me a bit

The table that just fixed my planform is exact only *conditional* on three numbers I did not derive: the ballast fraction $\beta = 0.4$ (a guess pending the CG budget), the fullness factor $K = 0.685$ (borrowed from an airfoil family I will not use), and the trim $C_L = 0.6$ (a stand-in for a min-sink optimum I cannot compute until I have picked the airfoil that this very chain is supposed to help me pick). I am making a 'final' geometric decision on top of three admitted approximations, then selecting an airfoil, then coming back to check whether the numbers I used deserved the trust I gave them.

The resolution is that this is not sloppiness, it is the only way the problem can be solved at all. Weight needs geometry and geometry needs weight; trim C_L needs the airfoil and the airfoil selection needs trim C_L . Every aircraft sizing process has these loops, and nobody solves them simultaneously: you cut each loop with a placeholder, close the chain, and then *verify convergence*: replace each placeholder with its computed value and check that no decision changes. The reason this works here is that all three placeholders enter the flight condition only under square roots ($V \propto \sqrt{W/S}$, $V \propto 1/\sqrt{C_L}$), so even a 10-20 percent error in a placeholder moves the Reynolds numbers a few percent; far less than the gap between rows of the table, and less than my hot-wire tolerance will smear the geometry anyway. The loops *contract*. The discipline is not avoiding placeholders; it is (1) writing down explicitly which numbers are placeholders and which are derived, so the audit is possible, and (2) knowing the sensitivity of each one, so I know how wrong a placeholder is allowed to be before it invalidates a decision. If a revisit ever shows a decision *does* flip, the placeholder was load-bearing and the loop gets iterated once more; minor tweaks, not a redesign, precisely because the couplings are weak.

1.5.3 Selecting my airfoil

The planform in 1.5.4 was closed against three placeholders, one of which was the trim $C_L = 0.6$ that I flagged as a stand-in for a min-sink optimum I could

not compute *until the airfoil was chosen*. That is the loop I now have to cut: the airfoil selection wants the station Reynolds numbers, which the planform already gives me ($Re_{root} \approx 70k$, $Re_{tip} \approx 42k$), but the min-sink polar that would let me rank candidates rigorously wants the airfoil I am trying to pick. Rather than stall on the circularity, I am doing to the airfoil exactly what 1.5.5 prescribes for every other coupled quantity: cut the loop with a defensible placeholder, start building, and verify convergence afterward.

The provisional pick: HS 3.0/8.0B. I adopt the HS 3.0/8.0B reflexed section as my day-one template. It is a purpose-built tailless/flying-wing section, so the reflex I need for tailless trim (the nose-up C_{m0} that enters *outside* the $\Gamma(y)$ analysis) is baked into the geometry rather than something I have to engineer. Its 8% thickness matches the t/c the sheet stock already forced on me in 1.5, so the section drops onto my root and tip stations without reopening the thickness or Reynolds decisions. And on a hot wire the reflex is free: it is inherited by every ruled station between the two templates. At a first pass, this is the candidate that best maps onto the constraints the tooling and material already fixed.

Why I am committing before the analysis. The honest reason is that the templates are the long pole in actually cutting a wing, and I wanted to begin fabricating rather than let the airfoil decision block the whole project. The *defensible* reason is that this is cheap to be wrong about. A template is a piece of plywood; if the analysis below turns the HS 3.0/8.0B out to be a poor choice at my Reynolds numbers, the cost of the mistake is one afternoon and a second pair of templates, not a redesign. This is the same sensitivity argument as 1.5.5: I commit where a wrong guess is cheap to revisit, and I hold off only where a wrong guess is load bearing.

What the analysis still owes me. Picking the section for fabrication is not the same as validating it, and the following checks are what convert this provisional choice into a justified one (or send me back to the plywood):

- **Polar at station Reynolds numbers.** Pull the c_l - c_d and c_l - α polars for HS 3.0/8.0B at $Re \approx 42k$ (tip) and $\approx 70k$ (root), and confirm the tip actually clears the laminar-bubble regime I was worried about in 1.5.4 rather than merely clearing the Re floor on paper.
- **Recover the real trim C_L .** Find the C_L that maximizes $C_L^{3/2}/C_D$ for this section and feed it back into the λ chain, checking that the min-sink C_L lands near the 0.6 placeholder. If it moves the tip Reynolds number less than the gap between rows of the 1.5.4 table, the taper choice stands.
- **Confirm C_{m0} and the trim budget.** Verify the section's nose-up moment is enough to trim the wing at the chosen sweep and washout without demanding more washout than one clean linear cut per panel can deliver.
- **Replace the borrowed fullness factor.** Swap the placeholder $K = 0.685$ (a NACA 4-digit number) for the HS 3.0/8.0B's true area factor

and re-check W/S , closing the last of the three 1.5.5 placeholders.

For now the decision I am acting on is simply this: HS 3.0/8.0B is the template I am cutting, and the subsections above are the audit that this choice has to survive.